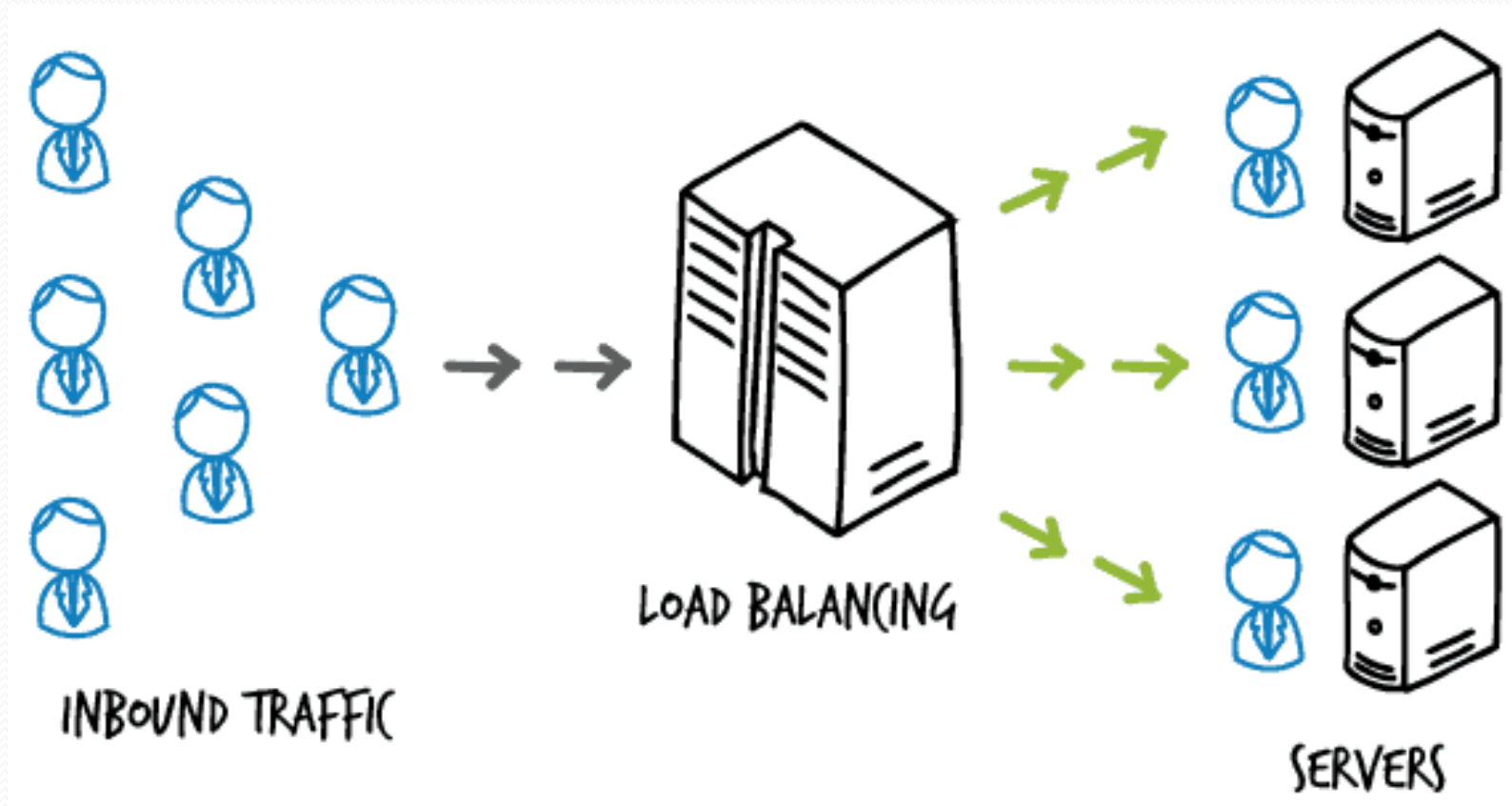


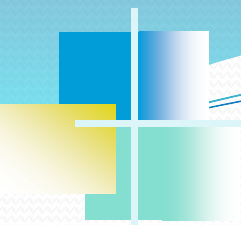


Load Balancing



SOURCES: Configuring Advanced Windows Server® 2012 Services

CCNA Data Center DCICT 640-916 Official Cert Guide.



Introduction

- Load balancing is a term that describes a method to distribute incoming socket connections to different servers. It's not distributed computing, where jobs are broken up into a series of sub-jobs, so each server does a fraction of the overall work. It's not that at all. Rather, incoming socket connections are spread out to different servers. Each incoming connection will communicate with the node it was delegated to, and the entire interaction will occur there. Each node is not aware of the other nodes existence.



Introduction

- In computing, load balancing improves the distribution of workloads across multiple computing resources, such as computers, a computer cluster, network links, central processing units, or disk drives. **Load balancing aims to optimize resource use, maximize throughput, minimize response time, and avoid overload of any single resource.** Using multiple components with load balancing instead of a single component may increase **reliability** and **availability** through **redundancy**. Load balancing usually involves dedicated **software** or **hardware**, such as a multilayer switch or a Domain Name System server process.



Why load balancing is employed

- There are at least two reasons why load balancing is employed:
 - ❑ **The required capacity is too large for a single machine.** When running processes that consume a large amount of system resources (e.g. CPU and memory), it often makes sense to employ multiple servers to distribute the work instead of constantly adding capacity to a single server. In plenty of cases, it's not even possible to allocate enough memory or CPU to a single machine to handle all of the work! Load balancing across multiple servers makes it possible to host high traffic websites or run complex data processing jobs that demand more resources than a single server can deliver.



why load balancing is employed

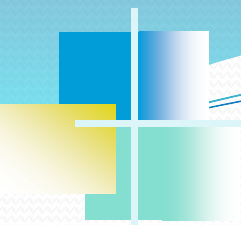
❑ Looking for more reliability and flexibility in a solution deployment.

Even if you *could* run an entire server application on a single server, it may not be a good idea. Load balancing can increase reliability by providing many servers able to do the **same job**. If one server becomes **unavailable**, the others can simply pick up the additional work until a new server comes online. **Software updates** become easier since a server can simply be taken out of the load balancing pool when a patch or reboot is necessary. Load balancing gives system administrators more flexibility in maintaining servers without negatively impacting the application as a whole.



Redundancy

- ❑ Servers crash, this is the rule, not the exception. Your architecture should be devised in a way to reduce or **eliminate single points of failure (SPOF)**. Load balancing a cluster of servers that perform the same role provides room for a server to be taken out manually for maintenance tasks, without taking down the system. You can also withstand a server crashing. This is called High Availability, or HA for short. Load balancing is a tactic that assists with High Availability, but is not High Availability by itself. To achieve high availability, you need automated monitoring that checks the status of the applications in your cluster, and automates taking servers out of rotation, in response to failure detected. These tools are often bundled into Load Balancing software and appliances, but sometimes need to be programmed independently



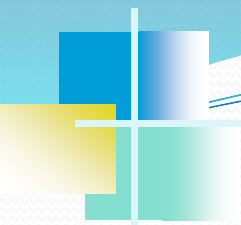
Network Load Balancing

- NLB is a scalable, high availability feature that you can install on all editions of Windows Server 2012. A scalable technology is one where you can add additional components (in this case additional cluster nodes) to meet increasing demand. A node in a Windows Server 2012 NLB cluster is a computer, either **physical** or **virtual**, that is running the Windows Server 2012 operating system.



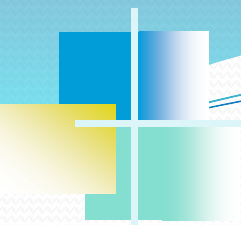
Network Load Balancing

- Windows Server 2012 NLB clusters can have between 2 and 32 nodes. When you create an NLB cluster, it creates a **virtual network address** and **virtual network adapter**. The virtual network adapter has an IP address and a media access control (MAC) address. Network traffic to this address is distributed evenly across the nodes in the cluster. In a basic NLB configuration, each node in an NLB cluster will service requests at a rate that is approximately equal to that of all other nodes in the cluster. When an NLB cluster receives a request, it will forward that request to the node that is currently **least utilized**. You can configure NLB to preference some nodes over others.



Network Load Balancing

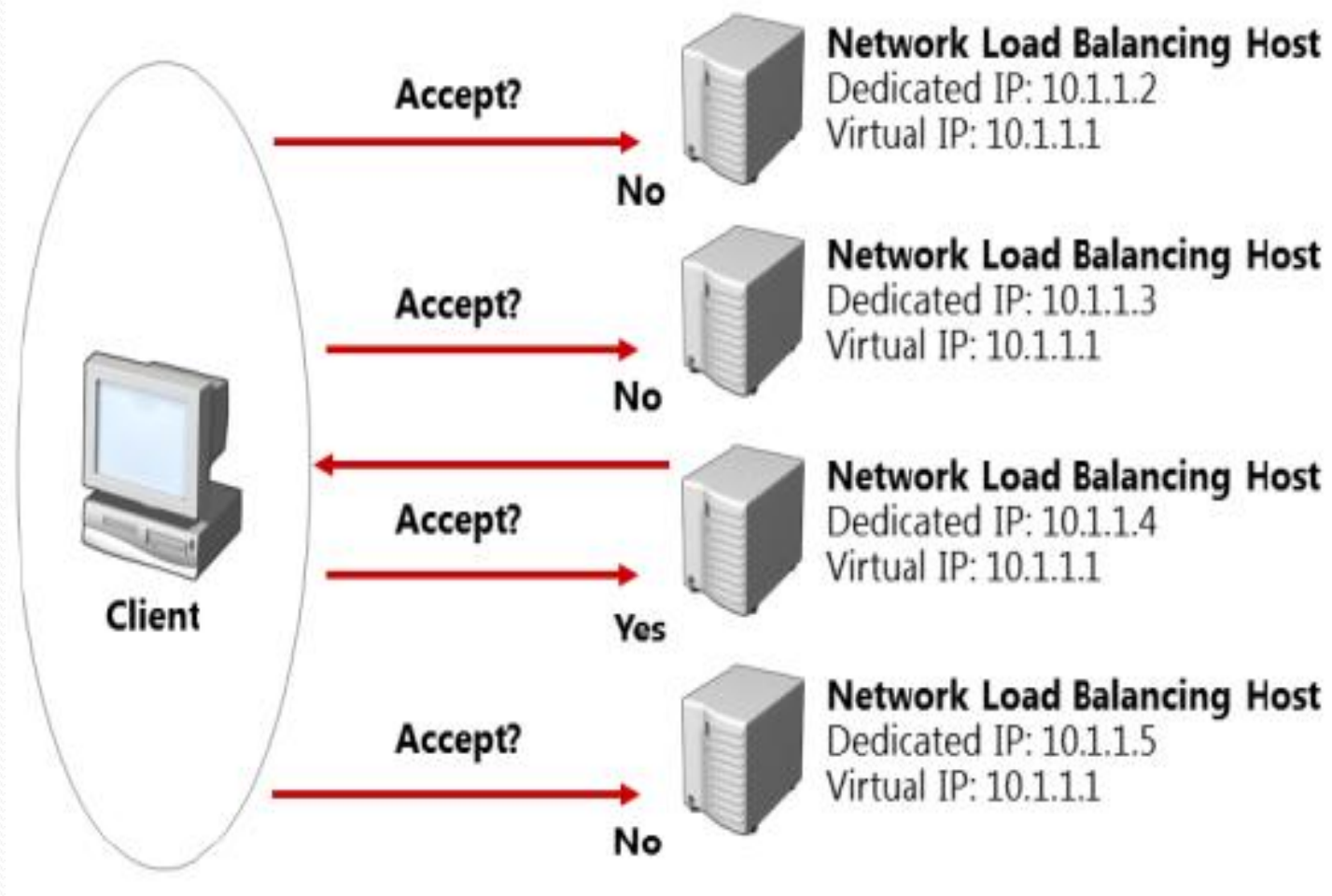
- **NLB is failure-aware.** This means that if one of the nodes in the NLB cluster goes offline, requests will no longer be forwarded to that node, but other nodes in the cluster will continue to accept requests. When the failed node returns to service, incoming requests will be redirected until traffic is balanced across all nodes in the cluster.



How NLB Works

- When you configure an application to use NLB, clients address the application using the **NLB cluster address** rather than the address of nodes that participate in the NLB cluster. The NLB cluster address is a **virtual address** that is shared between the hosts in the NLB cluster.
- NLB directs traffic in the following manner: All hosts in the NLB cluster receive the incoming traffic, but only one node in the cluster, which is **determined through the NLB process**, will accept that traffic. All other nodes in the NLB cluster will drop the traffic.

How NLB Works





How NLB Works

- Which node in the NLB cluster accepts the traffic depends on the configuration of **port rules** and **affinity settings**. Through these settings, you can determine if traffic that uses a particular port and protocol will be accepted by a particular node, or whether any node in the cluster will be able to accept and respond.
- NLB also sends traffic to nodes based on **current node utilization**. New traffic is directed to nodes that are being least utilized. For example, if you have a four node cluster where three of the nodes are responding to requests from 10 clients and one node is responding to requests from 5 clients, the node that has fewer clients will receive more incoming traffic until utilization is more evenly balanced across the nodes.

How NLB Works with Server Failures and Recovery

- NLB is able to detect the failure of cluster nodes. When a cluster node is in a failed state, it is removed from the cluster, and the cluster does not direct new traffic to the node. Failure is detected by using **heartbeats**. NLB cluster **heartbeats are transmitted every second between nodes in a cluster**. A node is automatically removed from a NLB cluster if **it misses five consecutive heartbeats**. Heartbeats are transmitted over a network that is usually different from the network that the client uses to access the cluster. When a node is added or removed from a cluster, a process known as **convergence occurs**. Convergence allows the cluster to determine its current configuration.



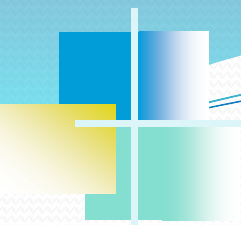
Deployment Requirements for NLB

- **NLB requires that all hosts in the NLB cluster reside on the same TCP/IP subnet.** Although TCP/IP subnets can be configured to span multiple geographic locations, NLB clusters are unlikely to achieve convergence successfully if the **latency between nodes exceeds 250 milliseconds** (ms). When you are designing geographically dispersed NLB clusters, you should instead choose to deploy an NLB cluster at each site, and then use Domain Name System (DNS) round robin to distribute traffic between sites.



Deployment Requirements for NLB

- **All network adapters within an NLB cluster must be configured as either unicast or multicast.** You cannot configure an NLB cluster where there is a mixture of unicast and multicast adapters. When using unicast mode, the network adapter must support changing its MAC address.
- You can only use TCP/IP protocol with network adapters that participate in NLB clusters. NLB supports IPv4 and IPv6. **The IP addresses of servers that participate in an NLB cluster must be static and must not be dynamically allocated.** When you install NLB, Dynamic Host Configuration Protocol (DHCP) is disabled on each interface that you configure to participate in the cluster.



Deployment Requirements for NLB

- All editions of Windows Server 2012 support NLB. Microsoft supports NLB clusters with nodes that are running different editions of Windows Server 2012. **However, as a best practice, NLB cluster nodes should be computers with similar hardware specifications, and that are running the same edition of the Windows Server 2012 operating system.**



What is Server Load Balancing?

- Server load balancing means to spread the workload among the servers hosting the same application content. **Server Load Balancer (SLB)** is a device that performs the load-balancing function. A load balancer receives the client request, analyzes the information in the request, and based on the load balancing **algorithm**, it divides the load appropriately between the servers. Load balancing can be done for client-to-server traffic, and for server-to-server traffic. Figure 19-1 shows an example of a typical load-balancing setup.

What is Server Load Balancing?

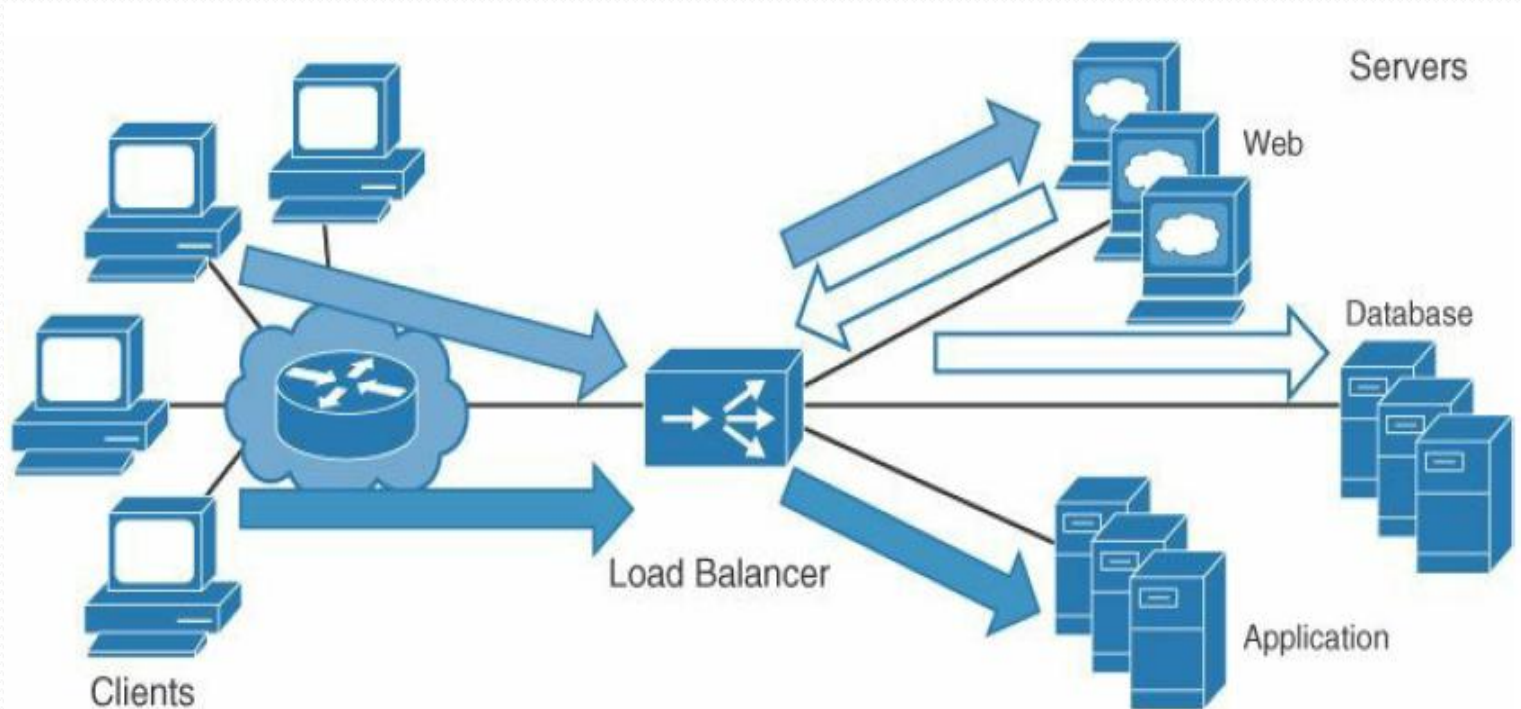


Figure 19-1 *Typical Server Load-Balancing Setup*



What is Server Load Balancing?

- When a load balancer receives multiple requests from clients for servers in a load-balanced farm, it distributes the requests evenly across all servers in the farm. Figure 19-2 shows an example where there are three servers in a data center. There is a load balancer between the clients and the servers, and this load balancer is configured for round-robin load balancing. The load balancer receives three requests from the clients. It divides these requests evenly among the three servers. Therefore, all three servers are equally busy serving the clients. The **Cisco Application Control Engine (ACE)** is an example of a network-based load balancer that can perform this task.

What is Server Load Balancing?

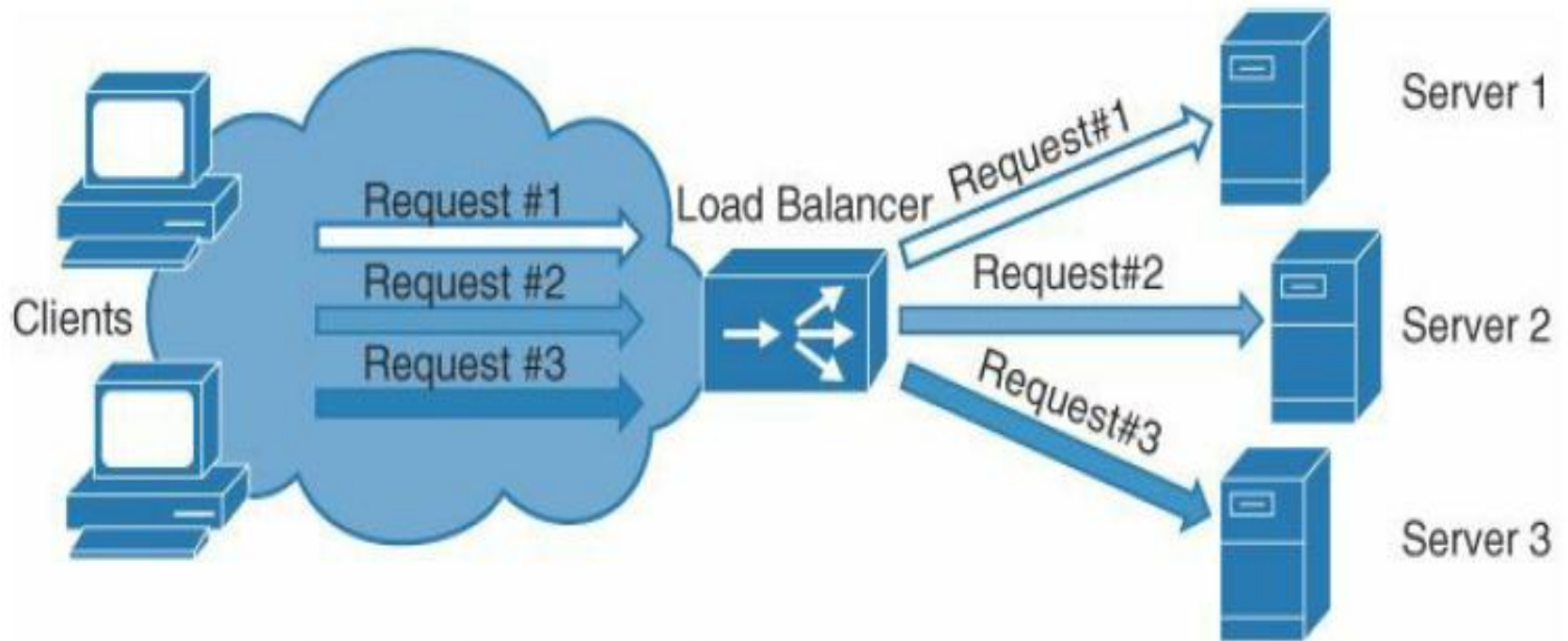


Figure 19-2 *Round-Robin Server Load Balancing*



Load balance algorithm

- **Round robin:** a batch of servers are programmed to handle in a rotating sequential manner. The algorithm assumes that each device is able to process the same number of requests and isn't able to account for active connections.
- **Weighted round robin:** Server are rated based on the relative amount of requests each is able to process. Those having higher capacities are sent more requests.
- **Least connections:** Requests are sent to the server having the fewest number of active connections, assuming all connections generate an equal amount of server load.



Load balance algorithm

- **Weight least connections:** Servers are rated based on their processing capabilities. Load is distributed according to both the relative capacity of the servers and the number of active connections on each one.
- **Source IP hash:** Combines the source and destination IP address in a request to generate a hash key, which is then designated to a specific server. This lets a dropped connection be returned to the same server originally handling it.
- **Least latency:** Makes a quick HTTP options request to backend server, and sends the request to the first server to answer.



Benefits of Server Load Balancing

Following are the benefits of the server load-balancing solution:

- **Scalability:** Scalability is one of the key benefits of a server load-balancing solution. As you have seen earlier, the load-balancing solution receives a client request for an application and divides the load among the servers running this application. **Based on your business requirements, you can start with a few servers and set up a load-balancing environment for them. As your business grows and the load on your existing servers increases, you can seamlessly add more servers to the solution.** The load balancer will include these new servers into the load-balancing calculation and start sharing the load with them. This approach allows high scalability because a large number of servers can be added to the solution without any outage.



Benefits of Server Load Balancing

- **Increased Capacity:** Server load balancing enables you to increase the capacity of your application so **it can deal with peak load from the client**. Typically, an application server can support only a limited number of connections from the client. A load-balancing solution helps you build an infrastructure that can **scale out** and support a greater number of connections; hence it provides increased capacity for your business applications. This increase in capacity can be done seamlessly by adding more servers in the solution.



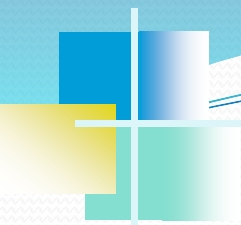
Benefits of Server Load Balancing

- **High Availability:** In a load-balanced server farm, many servers are serving the client requests. Therefore, **if one server in the farm goes down, the load balancer can shift the load to an alternate server.** The load balancer also performs a **health check** for each server in the server farm. If a server is not healthy it can be removed from the service to avoid interruption to the clients.
- **Disaster Recovery:** A global load-balancing solution enables you to put your servers across multiple sites. This feature is commonly used in disaster recovery (DR) solutions, where clients can continue accessing the service during the disaster **using the same server name.**



Benefits of Server Load Balancing

- **Security:** In a load-balancing environment, all the client requests are first received by a load balancer; therefore, the **load balancer can perform certain security checks before forwarding these requests to the servers.** One of the important security features that load balancers can apply is the access control list (ACL), where you can allow or deny traffic based on the information in the client request. Another example of a security task that load balancers typically perform is application protocol inspection, which helps to verify the protocol behavior and identify unwanted or malicious traffic that passes through the load balancer. You can define policies to accept or reject the packets to ensure the secure use of applications and services. Load balancers can also perform IP and TCP normalization, which protects against a variety of network attacks by performing general security checks, within IP and TCP headers.



Benefits of Server Load Balancing

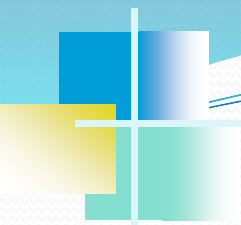
- **Operational Efficiency:** In a production data center, operators typically perform maintenance tasks on the servers. These tasks could be related to applications or hardware issues for which the server might need interruption. In a load-balancing solution, **the ability to add and delete servers without any outage provides operational efficiency**. You can bring down a server and perform maintenance without impacting the service.



Methods of Server Load Balancing

The following methods are used in the industry to perform server load balancing:

- **DNS-based load balancing:** This is an early method of performing server load balancing. In this method, DNS name resolution is used to distribute the load among the servers. DNS server A records are configured with multiple IP addresses for a name. Name resolution is done in a round-robin fashion, distributing the load among all the servers defined in DNS. There are several disadvantages of this approach:
 - ✓ DNS server **has no knowledge of server health**. It can resolve the name to an IP even if the server is down.
 - ✓ DNS caching works against the load balancing.
 - ✓ It supports only a very basic method of load balancing.



Methods of Server Load Balancing

- **IP Anycast:** A simple and effective way to perform load balancing of traffic between the geographic locations. In this method **IP routing techniques are used to redirect traffic to the server in the nearest data center location.** This is achieved by announcing the same IP subnet from multiple locations within a network. The routers then send the traffic to the preferred location based on their routing table.

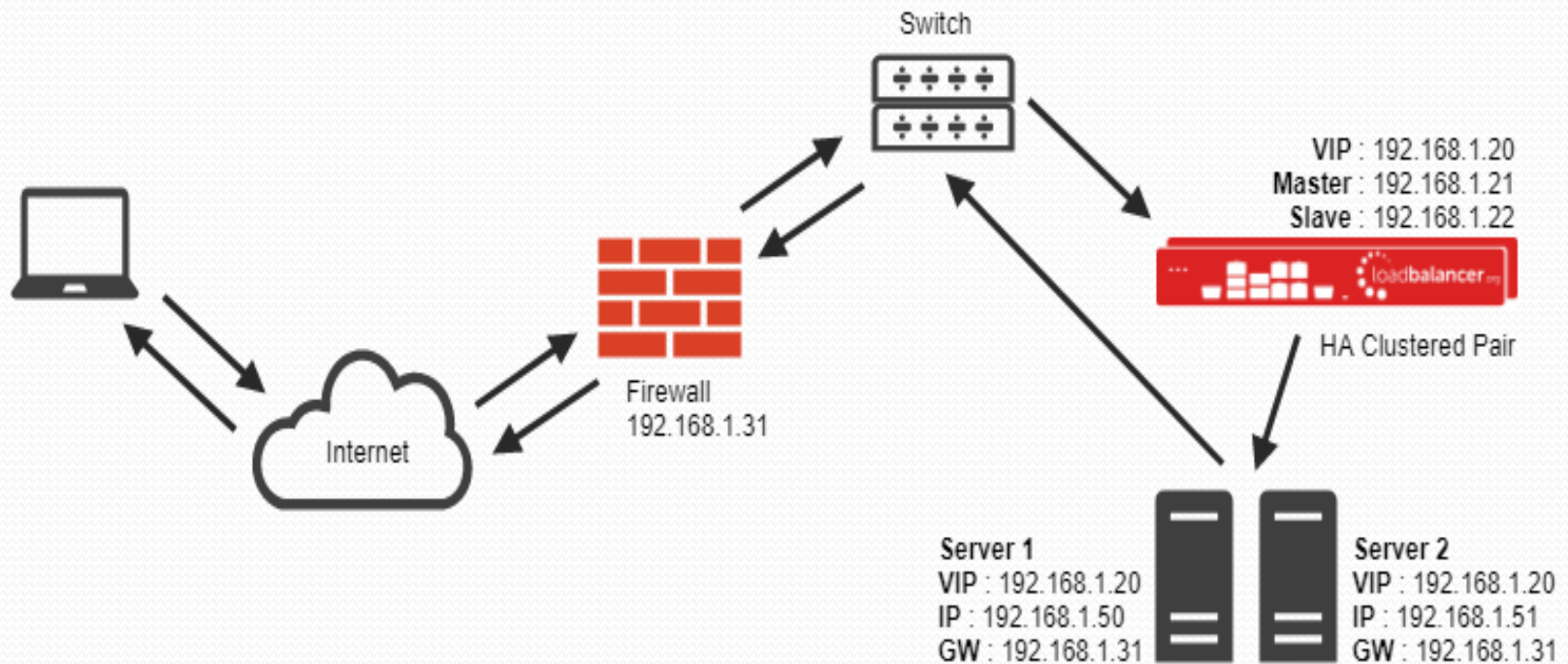


Methods of Server Load Balancing

- **Server-based load-balancing solutions:** In this solution, the load-balancing function is performed by the **server itself**. This can be achieved using features such as load-balancing server clusters or **special software that runs on the server** and performs the load balancing of traffic between the servers.
- **Network-based load-balancing solutions:** Server load balancing can also be achieved within the network. In this solution servers can perform the application function while the network can take care of the load balancing. This approach takes advantage of hardware acceleration within the network device; hence it improves the performance of the solution. **Network-based load balancing services are deployed at the aggregation layer of the data center**. These load balancers could be a **router** with the IOS SLB feature or a **specialized switch**, such as Cisco Application Control Engine (ACE).

Load Balance Type

- Network layer algorithms:



Load Balance Type

- Application layer algorithm:

Layer 7 Load Balancing

